Smart Personal Safety and Environmental Monitoring System Using ESP32

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Abstract*—To ensure human well-being, modern safety monitoring systems require efficient and real-time detection of hazardous environmental and physiological conditions. Traditional systems often rely on single-parameter monitoring, which can lead to limitation in their ability to respond effectively to multiple simultaneous risks. To address this limitation, this paper presents a smart multi-sensor-based personal safety and environmental monitoring system using an ESP32 microcontroller. The proposed system integrates an MQ-135 gas sensor for detecting harmful gases, a DHT22 sensor for monitoring temperature and humidity, and a MAX30102 sensor for real-time heart rate monitoring. By combining environmental and physiological parameters, the system provides an in-depth safety monitoring in real time. A priority-based decision-making algorithm is implemented to order to classify and respond to critical events, where gas leakage is given the highest priority due to its immediate hazardous nature, followed by abnormal heart rate conditions, elevated temperature levels, and humidity variations. Through visual indicators such as LEDs, an audible buzzer, and real-time status display on a 16×2 LCD module the system provides instant alerts. The use of the ESP32 platform enables efficient processing and reliable performance for real-time applications. By ensuring timely detection and response to multiple risk factors this integrated approach enhances the effectiveness of safety systems, making it suitable for household, industrial, and healthcare environments. The system gives immediate warnings through visual signals which include LEDs and an audible buzzer and a 16×2 LCD module which shows current system status. The ESP32 platform enables efficient processing which supports reliable real-time application performance. The integrated approach improves safety systems as it enables quick identification and response to various possible dangers which makes it appropriate for residential areas industrial facilities and medical settings. Keywords- Embedded systems, ESP32, safety monitoring, multi-sensor integration, gas detection, temperature and humidity sensing, heart rate monitoring, real-time systems

***Index Terms*—Embedded systems, ESP32, safety monitoring, gas detection, DHT22, MAX30102, real-time systems.**

I. Introduction

In recent times, there is a growing need for continuous safety monitoring, which has led to creating smart systems. They detect dangerous environmental and health conditions right away. Places like homes and job sites and medical areas often face many risks. These risks include hazardous gas leaks and unusual temperature and humidity levels and unnoticed changes in human health. Not catching these issues early can have serious effects, which proves why dependable and combined monitoring solutions are so important. Traditional safety systems usually watch only one parameter. This limits their use in changing and multi-risk settings. They often miss urgent situations as they can't fully explain the surrounding conditions. We can now make small systems on a budget with multiple sensors and live analysis with new embedded systems and IoT tech. But handling many inputs well and acting quickly in urgent cases is still tough. To fix these issues, this paper looks at a smart system for safety and monitoring using the ESP32 microcontroller. It includes a gas sensor to spot harmful gases and a temperature and humidity sensor for checking the environment. There's also a heart rate sensor for live health analysis. By combining environmental and health details, this system offers a seamless monitoring tool. The creation of a priority-based alert mechanism serves as a primary achievement for the proposed system because the design manages different danger levels well. According to the experts the highest ranking is given to the gas leakage because this danger presents a fast threat to the protection of people. Situations where heart rates are not normal follow this ranking because urgent health crises require fast attention. Environmental risks are identified when temperature changes happen, while humidity levels are usually seen as less important factors that influence long-term health. The priority-based alert mechanism provides a way for critical warnings to be sent quickly as the structured order prevents delays. The current information is used by the system to produce warnings, as is shown by visual signals, a loud sounder, and a visual screen unit. Researchers observe that the ESP32 platform allows for fast work and low energy use during continuous tracking tasks. This priority-based alert mechanism increases safety as different sensors and smart choices are combined together. The system is useful for many different actual situations because the design works efficiently in various working positions.

II. Literature Review

Existing monitoring systems, based on IoT, are usually concerned with a single parameter like air quality, temperature, humidity or physiological signals. Gas sensing systems employing MQ-series sensors are capable of detecting harmful gases in a contained location and generating alert signals accordingly. Environmental monitoring systems which utilize a digital sensor are able to monitor temperature and humidity of a desired location. Yet, these systems usually only allows single domain monitoring. Systems that use optical sensors for health monitoring can measure heart rate in real time and detect anomalies early on. However, these systems are usually not integrated into systems for monitoring environmental parameters. More recent methods have privileged multi-sensor integration for enhanced functionality. Many existing systems have no configuration to prioritize most critical conditions, which will lower responses in case of real use. The designed system overcomes these limitations by incorporating environmental and physiological monitoring along with a priority-based alert scheme capable of managing multiple risk conditions efficiently.

III. Proposed System

The system presents a smart multi-sensor safety solution which operates continuously to detect and respond to dangerous environmental and human health conditions. The system uses an ESP32 microcontroller as its primary component which handles data gathering and processing and emergency notification creation. The system uses three sensors which include a gas sensor that detects dangerous gases and a temperature and humidity sensor which monitors environmental conditions and a heart rate sensor that measures body functions. The ESP32 system performs real-time data processing to determine system operational status as the sensors transmit continuous data. The system uses a priority-based decision-making algorithm to manage multiple input streams through its operational system. The system assigns gas leakage detection as the top priority because it presents immediate threats and severe danger while medical emergencies show through abnormal heart rate detection. The system considers temperature changes as environmental hazards while it see humidity levels as minor risk factors. The system creates alerts through visual indicators and an audible buzzer and a display module which presents current system operational status. The system uses integrated methods to achieve precise detection which enables rapid response times that improve safety and operational dependability.

IV. System Architecture

The system architecture consists of three main layers: sensing, processing, and alerting. The sensing layer includes the gas sensor, temperature and humidity sensor, and heart rate sensor, which are joined with the ESP32 microcontroller. These sensors continuously monitor environmental and physiological parameters and transmit the collected data to the processing unit. The processing layer is handled by the ESP32, which reads sensor inputs, performs data analysis, and executes the priority-based logic to determine the severity of conditions. The decision-making process makes sure that critical events are identified and handled with higher precedence. The alerting layer comprises output components such as LEDs, a buzzer, and an LCD display. Depending on the detected condition, the system activates needed alerts, including visual signals, sound notifications, and real-time status display. This modular architecture enables efficient data flow, reliable operation, and easy scalability, making the system suitable for real-time safety monitoring applications.

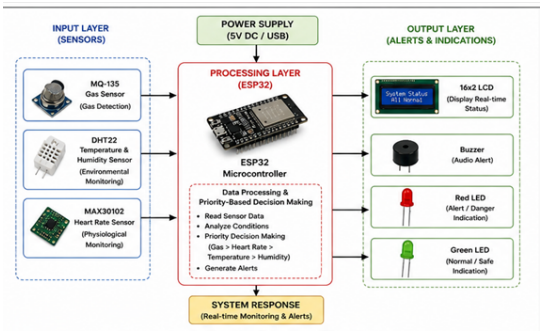


Fig. 1. Block Diagram

V. Methodology

The proposed system describes a methodology for real time sensing and processing, generating alerts using multiple sensors. The ESP32 microcontroller, the heart of this project connects to several environmental and physiological sensors. First, the gas sensor begins with a calibration stage to determine the basic value under normal conditions. The system reads the input data like gas sensor, temperature and humidity sensor and heart rate sensor during the operation. The data collected is processed in real time to evaluate the current state of the environment and user health. To adapt the multiple conditions, we implemented a priority decision-making algorithm. Gas leakage detection is given the top priority by the system on account of the hazard it presents. After this, abnormal heart rate conditions emerge, showing possible medical emergencies. Temperature variations are next considered after environmental risks and humidity levels are considered as secondary conditions. The evaluated conditions of the system activate the relevant alerts. LEDs are used for visual alarm, where red indicates abnormal and green indicates normal. A buzzer beeper will buzz in case of emergency alerts and through 16×2 LCD will be displayed. This method guarantees better care , supervision , constructive handling, and timely reactions to several risk situations, significantly increasing system reliability and efficiency.

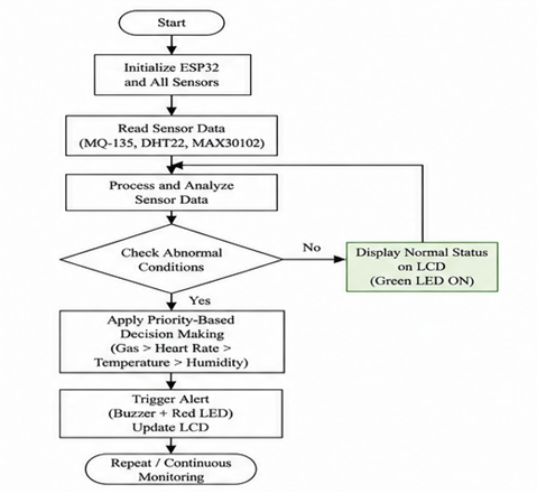


Fig. 2. Methodology

VI. Hardware Components

S. No.	COMPONENT	PURPOSE
1.	ESP32 Development Board (ESP32 DevKit V1)	Acts as the main controller of the system. It processes sensor data and controls all modules and alerts.
2.	DHT22 Temperature & Humidity Sensor	Measures ambient temperature and humidity for environmental monitoring.
3.	MQ-135 Gas Sensor Module	Detects harmful gases and air pollutants in the environment.
4.	MAX30102 Heart Rate Sensor	Measures heart rate (BPM) and helps in monitoring the user's health condition.
5.	16×2 LCD Display with I2C Module	Displays real-time data, system status, and alert messages to the user.
6.	Active Buzzer	Provides an audible alert during abnormal or critical conditions.
7.	Red LED	Indicates abnormal conditions or high-priority alerts.
8.	Green LED	Indicates normal operation and safe conditions.
9.	220Ω Resistors (2 Nos)	Limits current through LEDs to prevent damage and ensure proper operation.
10.	Breadboard	Provides a platform for easy prototyping and connections.
11.	Jumper Wires	Used for making connections between components on the breadboard.
12.	USB Cable	Provides power to the system and enables programming of the ESP32 board.

VII. Software Implementation

The proposed system's software is developed in Embedded C/C++ in the Arduino IDE for the ESP32 Microcontroller. The implementation highlights dependable real-time data collection, processing, and decision making with priority-based for safety monitoring application. All the input and output peripherals are configured when the system is initialized. Sensors interface and alert module are also configured. In normal environmental conditions, a calibration routine for the gas sensor is carried out to obtain a stable baseline reference. The DHT22 sensor is used to obtain temperature and humidity values of the environment and the heart rate sensor is used to get real-time functionality. The main logic runs continuously in a loop to continuously read, validate, and process the sensor value. A priority-based algorithm is implemented through which gas leakage is treated as the first and highest condition. Next is the abnormal heart rate condition then temperature and then humidity. According to evaluated condition of the system, alerts will be create using LEDs and buzzer. 16×2 LCD Module will display live status. To help keep the system working smoothly and reliably, the software has basic error handling and timing control. This makes sure of precision monitoring, effective response, and reliable performance and that qualifies the system for real-time safety-critical applications.

VIII. Results and Discussion

The design and implementation of the Smart Personal Safety and Environmental Monitoring System have been completed using the ESP32 development board along with gas, environmental and physiological sensors. This proposed monitoring system continuously sensed the required parameters and sent real time alerts if any abnormality was observed. The gas sensing module successfully sensed the variation of air quality or hazardous gas presence. In case of unsafe gas levels, the system issued alerts of highest priority level through buzzer, LEDs, and LCD display. The environmental sensing module accurately detected the temperature and humidity parameters and categorized them on different alert levels. Physiological heart rate monitoring module offered the capability of real-time monitoring of physiological conditions and generated an emergency alert in case of abnormal heart rates. Priority-based decision-making technique has been employed to handle all the abnormalities at once by considering the highest priority among them. LCD display showed the current status of system while buzzer and LED indicators produced alerts. The simulations results confirmed the proper functionality of sensor interfacing and alert generation mechanism. The presented system shows promising performance and low cost of implementation. Also, the proposed solution offers multi-sensor-based integration for real-time monitoring.

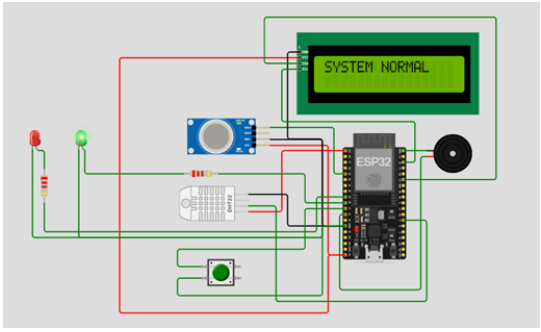


Fig. 3. Normal Operating Condition

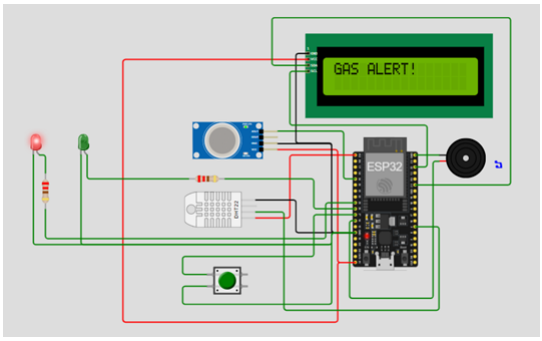


Fig. 4. Gas Leakage Detection

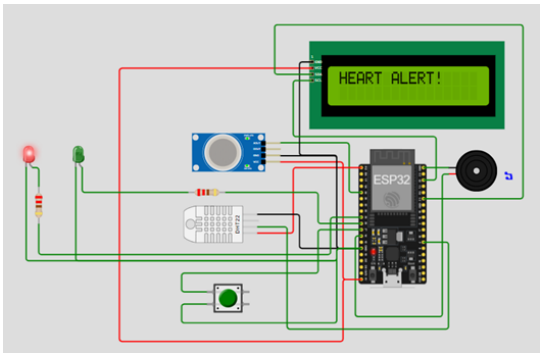


Fig. 5. Heart Rate Abnormality Detection

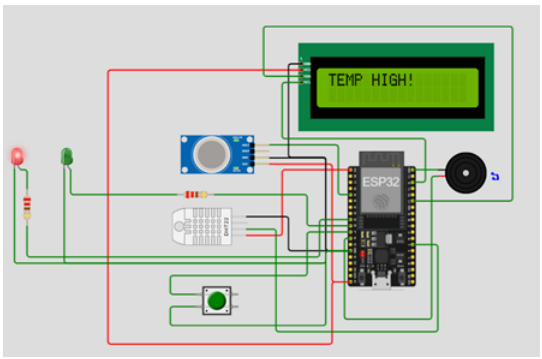


Fig. 6. a)Environmental Alert Condition

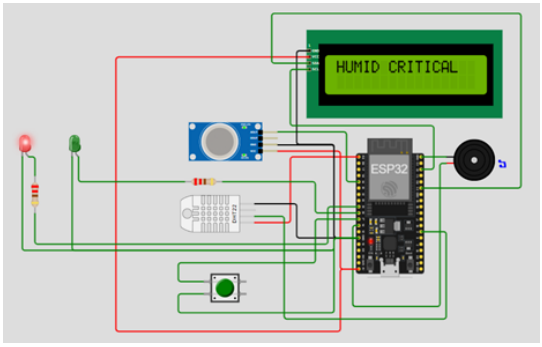


Fig. 7. b)Environmental Alert Condition

TABLE II. SYSTEM OUTPUT UNDER DIFFERENT CONDITIONS

Condition	LCD Output	Red LED	Green LED	Buzzer
Normal Condition	SYSTEM NORMAL	OFF	ON	OFF
Gas Leakage Detected	GAS ALERT!	ON	OFF	ON
Heart Rate Abnormal (Button Pressed)	HEART ALERT!	ON	OFF	ON
High Temperature	TEMP HIGH!	ON	OFF	ON
High Humidity Critical	HUMID CRITICAL	ON	OFF	ON

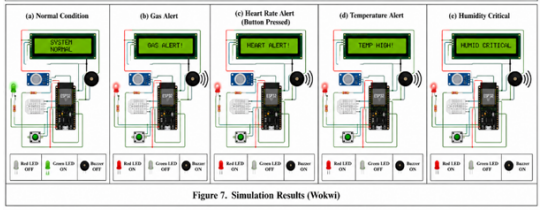


Figure 7. Simulation Results (Wokwi)

Fig. 8. Output Table

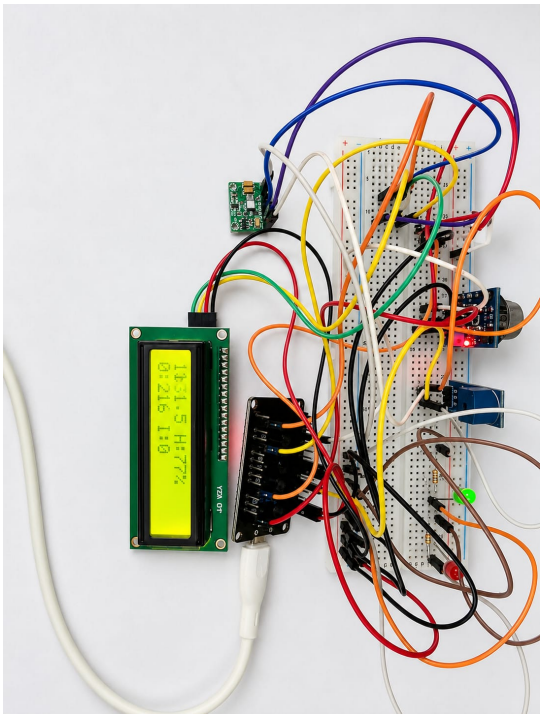


Fig. 9. Hardware

IX. Advantages

- Multi-sensor integrated safety monitoring system.
- Real-time detection of environmental and physiological abnormalities.
- Priority-based alert generation.
- Low-cost and portable design.
- Works independently without internet dependency.

X. Applications

- Industrial safety monitoring.
- Home gas leak detection.
- Elderly health monitoring.
- Healthcare monitoring systems.
- Indoor temperature and humidity monitoring.

XI. Conclusion

The developed smart personal safety and environmental monitoring system provides an efficient method for monitoring environmental and physiological parameters. By integrating gas detection, temperature and humidity sensing, and heart rate monitoring into a single embedded platform, the system improves safety and response time. The priority-based decision-making algorithm ensures that critical alerts are handled effectively.

XII. Future Scope

The system can be improved by adding IoT-based cloud connectivity for remote monitoring. A mobile application can be developed to receive real-time alerts. Additional sensors and AI-based prediction can also be included for improved decision-making.

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